Data-Driven Curriculum Optimizer

Academic Analysis Report | Generated 21 October 2026, 17:17

Student Performance

student_id	student_name	average_assessment_score	average_attendance	performance_category	risk_status
AID001	Student AID001	87.61	88.33	High	Low Risk
AID002	Student AID002	76.73	85.83	High	Low Risk
AID003	Student AID003	59.09	78.33	Average	Watchlist
AID004	Student AID004	89.22	89.17	High	Low Risk
AID005	Student AID005	89.27	89.17	High	Low Risk
AID006	Student AID006	71.37	80.83	Average	Low Risk
AID007	Student AID007	79.9	86.67	High	Low Risk
AID008	Student AID008	59.08	80.0	Average	Low Risk
AID009	Student AID009	63.01	79.17	Average	Low Risk
AID010	Student AID010	82.9	86.67	High	Low Risk
AID011	Student AID011	79.21	87.5	High	Low Risk
AID012	Student AID012	76.74	84.17	High	Low Risk
AID013	Student AID013	63.23	81.67	Average	Low Risk
AID014	Student AID014	64.31	81.67	Average	Low Risk
AID015	Student AID015	82.27	89.17	High	Low Risk
AID016	Student AID016	62.92	76.67	Average	Low Risk
AID017	Student AID017	87.16	88.33	High	Low Risk
AID018	Student AID018	65.38	80.83	Average	Low Risk
AID019	Student AID019	88.9	90.83	High	Low Risk
AID020	Student AID020	65.31	78.33	Average	Low Risk
CSE001	Student CSE001	77.63	85.83	High	Low Risk
CSE002	Student CSE002	78.73	89.17	High	Low Risk

student_id	student_name	average_assessment_score	average_attendance	performance_category	risk_status
CSE003	Student CSE003	60.74	80.83	Average	Low Risk
CSE004	Student CSE004	83.54	91.67	High	Low Risk
CSE005	Student CSE005	73.26	85.0	Average	Low Risk

Showing 25 of 60 rows. The full dataset is available in the CSV export.

Curriculum Effectiveness

course_id	course_name	average_assessment_score	average_attendance	effectiveness_score	effectiveness_status
AID101	Machine Learning	73.63	85.5	77.28	Effective
AID102	Data Analytics	79.67	83.75	80.67	Effective
AID103	Artificial Intelligence	70.74	83.25	75.29	Effective
CSE101	Data Structures	75.29	86.0	78.53	Effective
CSE102	Database Management Systems	71.33	84.75	75.63	Effective
CSE103	Computer Networks	68.34	85.0	74.2	Effective
ECE101	Digital Electronics	75.28	85.88	77.98	Effective
ECE102	Communication Systems	71.12	85.5	75.21	Effective
ECE103	Microprocessors	67.38	85.38	73.62	Effective

Learning Outcome Analysis

course_id	lo_code	course_name	target_attainment	calculated_attainment	attainment_status	priority
CSE101	LO1	Data Structures	70.0	75.29	Target Achieved	Normal
CSE101	LO2	Data Structures	70.0	75.29	Target Achieved	Normal
CSE102	LO1	Database Management Systems	70.0	71.33	Target Achieved	Normal
CSE102	LO2	Database Management Systems	70.0	71.33	Target Achieved	Normal
CSE103	LO1	Computer Networks	70.0	68.34	Below Target	Medium
CSE103	LO2	Computer Networks	70.0	68.34	Below Target	Medium
AID101	LO1	Machine Learning	70.0	73.63	Target Achieved	Normal
AID101	LO2	Machine Learning	70.0	73.63	Target Achieved	Normal
AID102	LO1	Data Analytics	70.0	79.67	Target Achieved	Normal
AID102	LO2	Data Analytics	70.0	79.67	Target Achieved	Normal
AID103	LO1	Artificial Intelligence	70.0	70.74	Target Achieved	Normal
AID103	LO2	Artificial Intelligence	70.0	70.74	Target Achieved	Normal
ECE101	LO1	Digital Electronics	70.0	75.28	Target Achieved	Normal
ECE101	LO2	Digital Electronics	70.0	75.28	Target Achieved	Normal
ECE102	LO1	Communication Systems	70.0	71.12	Target Achieved	Normal
ECE102	LO2	Communication Systems	70.0	71.12	Target Achieved	Normal
ECE103	LO1	Microprocessors	70.0	67.38	Below Target	Medium
ECE103	LO2	Microprocessors	70.0	67.38	Below Target	Medium

Recommendations

source	course_id	course_name	issue	priority	recommendation
Student Risk	AID102	Data Analytics	Students require additional academic attention	Medium	Provide targeted academic support and monitor stu
Student Watchlist	ALL	Multiple Courses	Students have course-level risk indicators	Medium	Monitor these students and provide course-specific
Student Risk	CSE102	Database Management Systems	Students require additional academic attention	Medium	Provide targeted academic support and monitor stu
Course Performance	CSE103	Computer Networks	Moderate assessment performance	Medium	Review challenging topics and provide additional le
Learning Outcome	CSE103	Computer Networks	LO1 below target attainment	Medium	Review learning activities and assessment alignme
Learning Outcome	CSE103	Computer Networks	LO2 below target attainment	Medium	Review learning activities and assessment alignme
Student Risk	CSE103	Computer Networks	Students require additional academic attention	Medium	Provide targeted academic support and monitor stu
Course Performance	ECE103	Microprocessors	Moderate assessment performance	Medium	Review challenging topics and provide additional le
Learning Outcome	ECE103	Microprocessors	LO1 below target attainment	Medium	Review learning activities and assessment alignme
Learning Outcome	ECE103	Microprocessors	LO2 below target attainment	Medium	Review learning activities and assessment alignme
Student Risk	ECE103	Microprocessors	Students require additional academic attention	Medium	Provide targeted academic support and monitor stu